Book V

Descents, Projections, and Certified Branch Legitimacy

Elements of Nested Fibrational Cosmology

Contents

V.0 Purpose

Book V is the descent and governance book of the canonical spine.

Book IV proved that the universal source object $\mathbf{U}$ exists, is bi-universal, and carries the canonical layered structure $T^* \hookrightarrow K^* \hookrightarrow \mathbf{U}$. The program now faces a new question: given that there is one universal source, how can it lawfully determine multiple downstream branches without collapsing them into arbitrary reinterpretations or permitting hidden imports?

Book V answers this question by defining and characterizing *lawful descent*. Its central theorem — the Universal Branch Legitimacy Theorem — gives a necessary and sufficient condition for a proposed downstream branch to count as a lawful descendant of $\mathbf{U}$. The five conditions are: explicit branch candidate intake, certified image of $\mathbf{U}$ as the endpoint datum, endpoint visibility, admission of a legitimacy witness, and exclusion of hidden-channel supplementation.

Book V is explicitly a *law-of-branchhood* book. It does not close any specific endpoint program. It does not prove Yang–Mills, Navier–Stokes, gravity, or any other specific result. Its purpose is to make the structure of lawful descent precise so that branch books cannot claim more than their declared descent machinery licenses.

1. Imports and Standing Rules

Imports. All certified results of Books I–IV. In particular: the bi-universal property of $\mathbf{U}$ (Book IV, Corollary IV.5.1), the canonical layered structure $T^* \hookrightarrow K^* \hookrightarrow \mathbf{U}$ (Book IV, Corollary IV.6.1), and the result that every lawful downstream realization factors through $\mathbf{U}$ (Book IV, Corollary IV.5.3).

Standing Rule 1.1 ([D]). A branch is legitimate only if its endpoint content is sourced through $\mathbf{U}$ and declared descent machinery.

Standing Rule 1.2 ([D]). No endpoint feature may be counted as branch structure unless it is branch-visible in certified terms.

Standing Rule 1.3 ([D]). A downstream claim may not import undeclared structure under the guise of interpretation, intuition, or endpoint convenience.

Standing Rule 1.4 ([D]). A branch must first be shown lawful before any branch-specific closure theorem has force.

Standing Rule 1.5 ([D]). Two branches are distinct only to the extent that certified endpoint-visible structure distinguishes them.

2. Branch Candidates and Descent Structure

Definition 2.1 ([D]). A *licensed target category* $\mathcal{C}$ is a mathematical category to which a faithful realization functor from POT may be directed, provided the functor is licensed by an explicit bridge theorem. Examples include: finite algebraic sector categories, operator/semigroup categories, and continuum realization categories (the last requiring the interface legitimacy of Book VI).

Definition 2.2 ([D]). A *realization functor* is a functor $F : \mathrm{POT} \to \mathcal{C}$ to a licensed target category that: preserves the POT-fibration structure; maps POT-morphisms to morphisms of $\mathcal{C}$; and is licensed by an explicit bridge theorem.

Definition 2.3 ([D]). A *licensed branch candidate* is a pair $(\mathcal{C}, F)$ consisting of a licensed target category $\mathcal{C}$ and a proposed realization functor $F : \mathrm{POT} \to \mathcal{C}$, together with an explicit declaration of the bridge theorem licensing F.

Definition 2.4 ([D]). The *endpoint datum* of a licensed branch candidate $(\mathcal{C}, F)$ is the object

$$E \;:=\; F(\mathbf{U}) \;\in\; \mathcal{C},$$

the image of the universal source object under the realization functor.

Definition 2.5 ([D]). A *branch projection* is the canonical morphism

$$p_F : \mathbf{U} \to F^{-1}(E)$$

expressing how $\mathbf{U}$ maps to the pre-image of the endpoint datum under F. At the POT level, it is the inclusion map $\eta_R : F_R \to \mathbf{U}$ post-composed with the realization functor.

Definition 2.6 ([D]). A *branch-visible observable* is any descendant observable φ on $E \in \mathcal{C}$ whose definability and value are invariant under replacement by source-equivalent data: if $\mathbf{U} \cong \mathbf{U}'$ as POT objects, then $F(\mathbf{U}) \cong F(\mathbf{U}')$, and φ assigns the same value to both.

Definition 2.7 ([D]). A *legitimacy witness* for a licensed branch candidate $(\mathcal{C}, F)$ is the certified evidence that:

(1) the endpoint datum $E = F(\mathbf{U})$ arises by certified descent from $\mathbf{U}$ through the declared realization functor F;
(2) E is branch-visible: its definition depends only on $\mathbf{U}$ and the declared functor F; and
(3) no undeclared supplementary structure beyond $\mathbf{U}$, F, and the declared branch-specific hypotheses is needed to define E.

Definition 2.8 ([D]). A *hidden channel* is any undeclared dependence of an endpoint claim on structure not visible through the declared branch projection and certified realization functor. Specifically, a hidden channel exists whenever:

(1) the endpoint E requires structure beyond what $F(\mathbf{U})$ carries; or
(2) the legitimacy of E implicitly assumes a result not in the declared import ledger.

Remark 2.9 ([R]). Hidden channels are the primary failure mode of ill-disciplined downstream programs. Common instances include: silently assuming a continuum structure not licensed by Book VI, implicitly importing a prior result not declared in the branch's import ledger, treating a conditional result as unconditional, and reading downstream physical interpretation back into source-level structure. Standing Rule 1.3 prohibits all such imports regardless of how they are framed.

3. Branch Legitimacy

Proposition 3.1 ([U]). (Branch Projection Theorem.) *For every licensed branch candidate $(\mathcal{C}, F)$, the endpoint datum $E = F(\mathbf{U})$ is well-defined in $\mathcal{C}$, and the branch projection is canonical: it depends only on $\mathbf{U}$ and the declared realization functor F.*

Proof. $\mathbf{U}$ exists and is unique up to unique POT-isomorphism (Book IV, Theorems IV.4.1 and IV.5.1). A realization functor $F : \mathrm{POT} \to \mathcal{C}$ maps POT-isomorphisms to isomorphisms in $\mathcal{C}$. Therefore $E = F(\mathbf{U})$ is well-defined up to isomorphism in $\mathcal{C}$, and the branch projection is canonical. □

Proposition 3.2 ([U]). *Under a licensed realization functor F, the endpoint datum $E = F(\mathbf{U})$ is branch-visible: its observational content is invariant under replacement by source-equivalent data.*

Proof. If $\mathbf{U} \cong \mathbf{U}'$ as POT objects (i.e., they are source-equivalent), then $F(\mathbf{U}) \cong F(\mathbf{U}')$ in $\mathcal{C}$ since F preserves isomorphisms. Therefore the observational content of E is invariant under source equivalence. Branch-visible observables on E inherit this invariance. □

Proposition 3.3 ([U]). (Legitimacy Witness Requirement.) *A proposed downstream branch has canonical endpoint force only if it admits a legitimacy witness.*

Proof. By Standing Rule 1.1 and Definition 2.7: canonical endpoint force requires that the endpoint datum depend only on the declared source descent and declared branch-specific hypotheses. Without a legitimacy witness certifying this dependency, the endpoint claim lacks the anti-smuggling guarantee required for canonical force. □

Proposition 3.4 ([U]). (Hidden Channel Exclusion.) *Any proposed downstream branch depending on a hidden channel is non-canonical. Specifically, if the endpoint datum E requires undeclared structure beyond $F(\mathbf{U})$ and the declared hypotheses, then no legitimacy witness exists.*

Proof. A legitimacy witness certifies (condition (3) of Definition 2.7) that no undeclared supplementary structure is needed. If a hidden channel exists (Definition 2.8), then condition (3) fails: there is undeclared structure influencing the endpoint datum. Therefore the legitimacy witness condition cannot be satisfied, and the branch claim is non-canonical. □

Corollary 3.5 ([U]). *Any descendant relying on hidden-channel supplementation is non-canonical.*

Remark 3.6 ([R]). Proposition 3.4 is the descendant-level anti-smuggling hinge of Book V. A branch becomes lawful not when it sounds plausible, but when its endpoint can be certified as source-descended without hidden supplementation. The three conditions of the legitimacy witness — certified descent, branch visibility, and no undeclared supplementation — together constitute the canonical test for branchhood.

4. Distinctness and Collapse Discipline

Lawful descent does not by itself decide when two branches are genuinely distinct. Book V must distinguish route-difference from endpoint-difference.

Definition 4.1 ([D]). Two branch projections p_F and $p_{F'}$ are *path-equivalent* if they arise from the same declared realization functor $F = F'$ via the same certified descent route from $\mathbf{U}$.

Definition 4.2 ([D]). Two branch projections p_F and $p_{F'}$ are *endpoint-equivalent* if their endpoint data agree up to the native equivalence of their target categories: $F(\mathbf{U}) \cong F'(\mathbf{U})$ in the relevant licensed category.

Definition 4.3 ([D]). A *collapse case* occurs when two proposed descendants are both path-equivalent and endpoint-equivalent.

Definition 4.4 ([D]). A *confluent distinctness case* occurs when two descendants are path-distinct but endpoint-equivalent: they arrive at the same certified endpoint structure via genuinely different certified routes.

Definition 4.5 ([D]). A *fully distinct case* occurs when two descendants differ both by path and by endpoint datum.

Theorem 4.6 ([U]). (Collapse Criterion.) *If two proposed descendants of* $\mathbf{U}$ *are path-equivalent and endpoint-equivalent, then they do not constitute distinct lawful branches.*

Proof. If the descendants share the same certified descent route and terminate in equivalent endpoint data, they differ neither by source path nor by target content. By Standing Rule 1.5, two branches are distinct only to the extent that certified endpoint-visible structure distinguishes them. Since no such distinction is present, the two proposed descendants are canonically the same branch. □

Corollary 4.7 ([U]). *Path multiplicity without endpoint distinction does not by itself generate new canonical branches.*

Corollary 4.8 ([U]). *Endpoint redescription without source-distinct descent does not generate new canonical branches.*

Remark 4.9 ([R]). Corollaries 4.7 and 4.8 block the two main failure modes of branch inflation: constructing a "new branch" by simply relabeling the same descent path, or repackaging the same endpoint datum with different notation. Both are excluded by the Collapse Criterion.

But collapse is not the only possibility. Distinct lawful paths may converge at the same endpoint without becoming identical as routes.

Definition 4.10 ([D]). A *diamond confluence configuration* is a pair of lawful descent paths from $\mathbf{U}$ into the same licensed target category whose endpoint data agree up to the native equivalence of that category.

Theorem 4.11 ([U]). (Diamond Confluence Principle.) *If two lawful descent paths from* $\mathbf{U}$ *into a declared target category are endpoint-equivalent, then they are canonically confluent at the endpoint, even when they remain path-distinct.*

PROOF. Endpoint-equivalence identifies the terminal descendant structure up to the target category's native equivalence. Path-distinctness records that the certified descent routes are different. Both facts can hold simultaneously: the endpoint-equivalence does not force the routes themselves to coincide. Therefore confluence at the endpoint does not require collapse of the routes. Two route-distinct but endpoint-equivalent descendants remain canonically meaningful as distinct paths to the same endpoint. □

COROLLARY 4.12 ([U]). *Confluence at the endpoint does not erase lawful path distinction.*

COROLLARY 4.13 ([U]). *Distinct descent paths may remain mathematically meaningful even when they terminate in the same certified endpoint structure.*

REMARK 4.14 ([R]). Taken together, the Collapse Criterion and the Diamond Confluence Principle say: Book V permits lawful multiplicity (distinct routes to a common endpoint) while forbidding arbitrary branch inflation (calling the same thing new merely by redescription). This is the correct balance.

5. The Universal Branch Legitimacy Theorem

The previous sections establish all five conditions needed for the governing theorem of this book.

THEOREM 5.1 ([U]). (Universal Branch Legitimacy Theorem.) *A proposed downstream branch $(\mathcal{C}, F, E)$ is lawful if and only if all five of the following conditions hold:*

(1) Licensed intake*: $(\mathcal{C}, F)$ is a licensed branch candidate with explicitly declared realization functor.*
(2) Certified source descent*: the endpoint datum $E = F(\mathbf{U})$ arises as the certified image of the universal source object $\mathbf{U}$ under the declared realization functor.*
(3) Endpoint visibility*: E is branch-visible — its definition and value are invariant under source equivalence.*
(4) Legitimacy witness*: a legitimacy witness for $(\mathcal{C}, F)$ exists, certifying that E depends only on $\mathbf{U}$, the declared functor F, and the declared branch-specific hypotheses.*
(5) No hidden-channel supplementation*: the endpoint claim requires no undeclared structure beyond what conditions* (1)–(4) *license.*

PROOF. *Necessity.* Each condition is necessary for canonical branchhood:

- Condition (1) is necessary by Definition 2.3: a branch must have a declared and licensed descent mechanism.
- Condition (2) is necessary by Book IV, Corollary IV.5.3: every lawful downstream realization factors through $\mathbf{U}$.
- Condition (3) is necessary by Standing Rule 1.2: a branch endpoint must be branch-visible.
- Condition (4) is necessary by Proposition 3.3: canonical force requires a legitimacy witness.
- Condition (5) is necessary by Proposition 3.4: any hidden channel invalidates the legitimacy witness.

Sufficiency. Conversely, conditions (1)–(5) are jointly sufficient:

- Condition (1) provides the declared descent structure.
- Condition (2) ensures the endpoint is sourced from $\mathbf{U}$.
- Condition (3) ensures the endpoint is canonically meaningful.
- Condition (4) certifies the dependency structure is free of smuggling.
- Condition (5) confirms no undeclared supplementation is present.

Once all five conditions are met, the descendant is fully source-certified, endpoint-visible, and free of hidden support. No further conditions are needed for the branch to count as a lawful descendant. □

Corollary 5.2 ([U]). *Every derived branch book must begin by declaring:*

(1) *its licensed target category* $\mathcal{C}$*,*
(2) *its realization functor* $F : \mathrm{POT} \to \mathcal{C}$ *and the bridge theorem licensing it,*
(3) *its endpoint datum* $E = F(\mathbf{U})$*,*
(4) *the native equivalence of* $\mathcal{C}$ *determining when two endpoint data coincide, and*
(5) *its legitimacy witness.*

A branch book that fails to supply these five items is not canonically admissible under Book V.

Proof. The five items correspond precisely to the five conditions of Theorem 5.1. Admissibility requires all five; omitting any one leaves the branch without canonical force. □

Corollary 5.3 ([U]). *A branch may be mathematically serious and canonically admissible even when its closure remains open, provided its target, descent law, and failure frontier are canonically declared.*

Proof. The Universal Branch Legitimacy Theorem certifies that a branch is *lawful*, not that it is *closed*. Lawfulness is a structural condition about source descent and hidden-channel exclusion. Closure is a further mathematical achievement requiring additional branch-specific theorems. A lawful but unclosed branch with an explicitly declared failure frontier satisfies all conditions of Theorem 5.1 and retains full canonical standing. □

Scholium 5.4 ([R]). Book V defines lawful descendant structure. It does not yet determine the exact force of descendant claims, nor does it license continuum language within them. The force of a branch claim is governed by Book VII. The licensing of continuum and differential language within a branch is governed by Book VI. Both of those books are downstream of Book V: they presuppose that a branch is lawful before governing how its claims should be stated.

6. Screened Sector as Canonical Endpoint Datum

This section records the Screening Inevitability Chain (Holding BIN CROSS, v7.35 §11791) and the canonical identification of the screened sector as a branch-visible endpoint datum satisfying Book V discipline.

PROPOSITION 6.1 ([C]). (τ-Channel Ceiling Bound.) [Conditional on $K_0 = 7$, LS-2, Phase-A.] *The number of independent boundary-channel parameters consistent with Phase-A stability is bounded above by $\tau(7) = 4$ (three d_{S_v}-block directions plus one gauge-invariant direction). Any observable family with more than $\tau(7)$ independent boundary-channel parameters violates Phase-A stability.*

PROOF. By the self-consistency fixed-point theorem (dream_paper_v38 §6.3, Theorem 6.5): the number of independent boundary parameters consistent with Phase-A is $\tau(p) \leq 5$ for all primes (the Avoidance-Cycle Exclusion theorem caps τ), and for $p = 7$ specifically $\tau(7) = 4$ (verified in §6.2). Any observable family exceeding this bound cannot arise from Phase-A data. □ □

LEMMA 6.2 ([C]). (PASS + LS-2 Forbid Hidden Decoupled Near-Resonances.) [Conditional on PASS (Thm. **??**, GR branch), LS-2 (Book II).] *Under PASS and LS-2, no decoupled near-resonant boundary mode can persist in the observable family without being captured by the screened sector W_A: any mode that commutes with the full G-action and survives LS-2 stabilization is already in W_A.*

PROOF. By PASS (Theorem **??**): every backbone observable in W_A commutes with G. Conversely, any G-commuting observable not already in W_A would define an additional G-invariant sector not accounted for in the d_{S_v}-block decomposition. By the τ-channel ceiling (Proposition 6.1), no such additional sector exists at $K_0 = 7$. Under LS-2, stabilized local observables are determined by collar type; any near-resonant mode that is LS-2 stable and G-commuting is already captured by the $K_0 = 7$ collar type structure and hence in W_A. □ □

THEOREM 6.3 ([C]). (Screened Sector as Canonical Branch-Visible Endpoint Datum.) [Conditional on PASS, LS-2, Phase-A, $K_0 = 7$.] *The screened sector $W_A = \ker(T_\partial - \lambda_{\max} I)^\perp$ is a canonical branch-visible endpoint datum in the sense of Definition 2.4:*

(1) Source-descended*: W_A is determined by T_∂ and the canonical screening rule, both source-descended (Lemma T3.1a, YM Branch).*
(2) Branch-visible*: W_A is the declared endpoint of the YM, NS, and SM branches — the screened observable algebra acts on W_A and no hidden channel contributes (Lemma 6.2).*
(3) Sector-unique*: W_A is the unique complement of the dominant eigenspace of T_∂; no alternative screened sector exists at the same τ-level (Proposition 6.1).*
(4) Book V compliant*: W_A arises as the certified image $F(\mathbf{U})|_{W_A}$ of the universal completion $\mathbf{U}$ through the canonical screening functor, satisfying the five UBLT conditions of Definition 2.4.*

PROOF. (1) Lemma T3.1a (YM Branch: canonical screened-sector definability): W_A is determined by $(\Sigma_\partial, T_\partial)$ alone, both source-descended. (2) By Lemma 6.2: all G-commuting LS-2-stable observables are in W_A; no hidden channel. (3) By Proposition 6.1: $\tau(7) = 4$ caps the number of independent sectors; W_A is the unique one matching the declared Phase-A architecture. (4) The screening functor $\Pi_A : \mathbf{U} \to W_A$ (the canonical orthogonal projection) is a morphism of the POT category (Book IV);

$W_A = F(\mathbf{U})|_{W_A}$ arises as the screened sub-object. The five UBLT conditions follow from (1)–(3) and the lawfulness of Π_A (Lemma T3.1c, YM Branch: bracket-compatible). □ □

Remark 6.4 ([R]). Theorem 6.3 closes the Holding entry "Screening Inevitability Chain" at the level of a conditional canonical theorem. The one remaining open hinge — Uniform Anti-Chasing (the analytic gap in the screening inevitability argument for non-abelian Gribov copies) — is not needed for the endpoint-datum identification theorem; it would be required for a stronger theorem asserting that *all* potential hidden channels are provably absent.

7. Boundary of Book V

What Book V proves.

(1) Lawful descent from $\mathbf{U}$ is governed by a five-condition biconditional: the Universal Branch Legitimacy Theorem.
(2) Every lawful branch must have a licensed target category, a declared realization functor, a certified endpoint datum, a legitimacy witness, and no hidden-channel supplementation.
(3) Path multiplicity without endpoint distinction does not generate new branches (Collapse Criterion).
(4) Distinct descent routes may converge at the same endpoint without collapsing (Diamond Confluence Principle).
(5) A branch is canonically admissible even when unclosed, provided its failure frontier is explicitly declared.

What Book V does not prove, and may not be assumed here.

(1) *Branch closures.* Book V certifies branchhood, not closure. Yang–Mills closure, Navier–Stokes regularity, and gravity closure all belong in derived branch books.
(2) *Continuum interface.* Book V does not license the use of differential, integral, geometric, or smooth language in branch books. That licensing belongs to Book VI.
(3) *Force of branch claims.* Book V does not determine whether a branch-level claim is unconditional, conditional, or bridge-dependent. That governance belongs to Book VII.
(4) *Physical endpoint identification.* The identification of any specific endpoint datum with a physical theory (gauge field, fluid, spacetime geometry) belongs in derived branch books and requires specific bridge theorems not proved here.

8. Handoff to Book VI and Book VII

Book V branches the canonical program in two directions.

Toward Book VI: When a lawful branch seeks to use continuum, differential, integral, or geometric language as an encoding of its endpoint structure, it must invoke Book VI's interface legitimacy machinery. No smooth language is licensed in a branch book without first establishing the relevant continuum interface regime through Book VI.

Toward Book VII: When a lawful branch seeks to state the exact force, scope, import discipline, and failure frontier of its claims, it must invoke Book VII's canonical closure schema. The governance of how branch claims may be promoted, cited, and combined belongs constitutionally to Book VII.

The handoff from Book V is therefore:

$$\begin{aligned}&\mathbf{U} \;\longrightarrow\; \text{licensed branch candidate} \;\longrightarrow\; \text{lawful branch projection}\\ &\longrightarrow \begin{cases}\text{Book VI: interface legitimacy,}\\ \text{Book VII: closure discipline.}\end{cases}\end{aligned}$$

Book V supplies the first three stages.

9. Dependency Ledger

Theorem	**St.**	**Depends on**	**Exports to**
3.1 Branch Projection Thm.	[U]	Bk. IV Thms. IV.4.1, IV.5.1	3.2, 5.1
3.2 Endpoint Visibility	[U]	3.1	5.1
3.3 Legitimacy Witness Req.	[U]	SR 1.1, Def. 2.7	3.4, 5.1
3.4 Hidden Channel Exclusion	[U]	Def. 2.8, 3.3	3.5, 5.1
4.6 Collapse Criterion	[U]	Defs. 4.1–4.3, SR 1.5	4.7, 4.8
4.11 Diamond Confluence Principle	[U]	Defs. 4.4, 4.10	4.12, 4.13
5.1 Universal Branch Legitimacy Thm.	[U]	3.1–3.4; Bks. I–IV	Books VI–VII; all branch books
5.2 Mandatory Declarations	[U]	5.1	Constitutional rule for all branch books
5.3 Open branch canonical	[U]	5.1	NS, YM, GR, SCC, A* branch books

Scholium 9.1 ([R]). The Universal Branch Legitimacy Theorem is the constitutional entry point for every derived branch book. No branch may begin its work without satisfying the five conditions of Theorem 5.1, declared explicitly at the opening of the branch manuscript. This requirement is not bureaucratic: each condition rules out a specific form of mathematical dishonesty. Condition (1) rules out unlicensed realization. Condition (2) rules out branches that do not actually descend from

$\mathbf{U}$. Condition (3) rules out unobservable endpoints. Condition (4) rules out undeclared dependencies. Condition (5) rules out hidden imports. A branch that satisfies all five conditions has genuinely earned its canonical status.

10. Coercive-Transport Source Theorem and ICDC Schema

THEOREM 10.1 ([C]). (Unified Coercive-Transport Source Theorem, BR.) [Conditional on $K_0 = 7$, the universal object $\mathbf{U} \in \mathrm{POT}$ (Book IV, Thm. **??**), and the controlled continuum realization (Book VI, Thm. **??**).] *There exists a single upstream coercive-transport structure on* $\mathbf{U}$ *with the following burden-routed content:*

(i) Source-level existence. *A coercive form* $\mathcal{B}(f,g) = \int_{\mathbf{U}} \langle \nabla_{\mathbf{U}} f, \nabla_{\mathbf{U}} g \rangle \, d\mu_{\mathrm{nl}}$ *is well-defined on* $\mathcal{H}_\infty = L^2(\mathbf{U}, \mu_{\mathrm{nl}})$*, with screened-sector lower bound* $\mathcal{B}(f,f) \geq c\|f\|^2$ *for some* $c > 0$*. This is the intrinsic content of the BR theorem.*

(ii) Yang–Mills specialization. *Under the realization and quantization inputs of the YM Branch, the source coercivity pushes forward to the Yang–Mills branch Hamiltonian estimate.*

(iii) Navier–Stokes specialization. *Under the realization and transport inputs of the NS Branch, the same source pushes forward to the NS branch floor/transport estimate.*

(iv) Structural Einstein specialization. *Under the geometric realization inputs of the GR Branch (KPO-3), the same source pushes forward to the EFE coercive structure.*

Accordingly, the YM, NS, and GR branch theorems are to be read as certified specializations of one upstream source theorem, not as independent foundations.

PROOF. The unconditional existence of the coercive form follows from Thm. **??** (Book IV), which provides both the transfer structure on $\mathbf{U}$ and the canonical transport measure μ_{nl}. Branch specialization under $K_0 = 7$ follows from the controlled continuum realization (Book VI, Thm. **??**), which licenses the push-forward of the coercive structure to the realized operators in each branch. No branch conclusion belongs to the unconditional BR theorem beyond this burden-routed transfer principle. (Attributed: dream_paper_v38.pdf Thm. 9.19.) □ □

THEOREM 10.2 ([C]). (Interface-Coherent Directed Compression, ICDC Schema.) [Conditional on $K_0 = 7$ and one-step inheritance.] *Let* $U_{n_0} \subseteq U_{n_0+1} \subseteq \cdots$ *be an admissible enlargement ladder from* $\mathbf{U}$*, and let* $(B_n)_{n \geq n_0}$ *be a branch family functorially extracted along this ladder. Assume:*

(1) One-step inheritance*: branch data transport lawfully from* B_n *to* B_{n+1}*;*

(2) Interface localization*: any failure of exact inheritance is confined to an explicit interface residue* ∂_n*;*

(3) Two-step transitivity*: corrections for* $n \to n+1 \to n+2$ *compose into a single accumulated residue* $\partial_{n,n+2}$*;*

(4) Source-image discipline*: every load-bearing datum of each* B_n *remains in the image of* $\mathbf{U}$ *with no new branch-local primitive introduced;*

(5) Admissible compression*: the accumulated interface residues are summably admissible or uniformly subordinate to a branch-specific coercive margin.*

Then the family admits a controlled scale-limit compression $B_\infty := \varinjlim_\partial B_n$ *in which the load-bearing branch data survive coherently across scale modulo explicit interface bookkeeping.*

PROOF. The one-step inheritance maps define the direct-system spine. Interface localization isolates all failure of exact transport in named residues. Two-step transitivity ensures residues assemble coherently. Source-image discipline blocks hidden downstream smuggling. Admissible compression yields a well-defined controlled limit object. (Attributed: dream_paper_v38.pdf Thm. 9.12.) □ □

REMARK 10.3 ([R]). **Branch-specific ICDC readings.** The ICDC schema specializes to three terminal branches:

- *Yang–Mills* ($\Theta_n = 1$, gap-subcritical): gap preservation holds whenever $E_n + B_n < g_\infty^{(n)}$;
- *Navier–Stokes* ($\Theta_n = \theta < 1$): obstruction contraction $C_{n+1}^{NS} \leq \theta C_n^{NS} + E_n + B_n$;
- *Gravity* ($\Theta_n = 1$, curvature-subcritical): domain extension holds when new interface contributes curvature defect strictly subordinate to the already-controlled gravity margin.

The common upstream law is the interface-relative stability functional $\mathcal{C}_n(U_n) = B_n^{\text{bulk}}(U_n) - \mathcal{I}_n(U_n)$; what changes branch-to-branch is only the terminal observable and closure criterion.